

Research Note 03

Longer Horizons, the Trend Band, and Where Traffic Forecasting Stops Being “Sunrise”

Traffic Forecasting Project

June 14, 2026

Abstract

Notes 01–02 showed one-week-ahead flow forecasting is effectively solved by a simple seasonal+holiday model ($R^2=0.949$): traffic, unlike asset prices, is “sunrise-like” – dominated by a deterministic cycle anchored to human schedules. This note asks what changes at *longer* horizons. We frame the signal as three frequency bands – *seasonal* (deterministic at any horizon), *noise* (bounded, averages out), and *trend/level* (forecast variance grows with horizon) – and answer, in *steady state*, at what horizon the problem becomes interesting. The transition is a **smooth onramp, not a sharp knee**: the “copy recent weeks” baseline degrades roughly linearly as its data goes stale, from a 4% gap to the floor at one week to 13% at four weeks and 25% at twelve. By the natural threshold – richer level/trend modeling recovering a holiday-scale margin – the problem becomes interesting at roughly **three to four weeks** out, and is unambiguously a trend problem by about eight weeks. The cause is structural: across 2017–2021 the weekly *shape* is frozen (cross-year correlation ≥ 0.99) while the *level* carries all inter-temporal action; longer horizons just expose more of that slow level band. (Regime breaks like COVID are the extreme tail of the same band – a one-off, not the everyday story.) We close on where richer statistical methods earn their keep.

Key findings

- **Frequency-band view:** forecast difficulty is set by which band dominates. Seasonal is deterministic; noise averages out; only the trend band has horizon-growing uncertainty.
- **A smooth onramp, no sharp knee.** The naive baseline’s gap to the no-staleness floor grows $\sim$ linearly: 4% (1 wk), 13% (4 wk), 25% (12 wk). The steady-state problem becomes “interesting” (richer modeling recovers a holiday-scale margin) at $H \approx 3\text{--}4$ **weeks**.
- **Each week of horizon ages the level anchor.** A horizon-independent annual anchor stays flat while trailing data goes stale, so its optimal blend weight rises $0.20 \rightarrow 0.39$ over 1–12 weeks – but it recovers only $\sim$ half the gap; the rest needs genuine trend modeling.
- **Seasonal shape is frozen; the level is everything.** 2017–2021 weekly-shape cross-year correlation ≥ 0.99 ; the network level moves only with slow growth and the COVID shock ($240 \rightarrow 205 \rightarrow 228$ veh/h). Longer horizons simply expose more of the slow level band.
- **Regime breaks are the tail, not the trend.** COVID spikes the one-week MAPE from $\sim 3.5\%$ to $\sim 60\%$ transiently – a vivid one-off, but the everyday long-horizon difficulty is the smooth level drift above.

1 A frequency-band view of forecast difficulty

Write flow as $y_{i,t} = \underbrace{T_{i,t}}_{\text{trend/level}} + \underbrace{S_{i,t}}_{\text{seasonal}} + \underbrace{\varepsilon_{i,t}}_{\text{noise}}$. At forecast lead h the three behave completely differently:

- **Seasonal** S is anchored to the clock/calendar (hence to the sun): deterministic given the date, so it contributes *zero* error at *any* h . This is why short-horizon traffic is “sunrise-like”.
- **Noise** ε (incidents, weather) is irreducible but bounded and, crucially, *averages down* under aggregation – the reason the weekly-aggregate MAPE is only 3.5% (Note 02).
- **Trend** T is a near-random-walk: its forecast variance *grows* with h . At one week it is negligible against S ; at months–years it is the whole game.

The contrast with asset prices follows: a price is almost *all* trend band (no seasonal anchor) and its structure is arbitrated away, so it is hard even at $h=1$. Traffic hands us a giant seasonal anchor that no one is incentivized to arbitrage away – so it is easy at short h , and only becomes price-like as the horizon erodes the anchor’s relative weight.

2 The onramp: at what horizon does it get interesting?

A subtle constraint governs multi-week horizons. At lead H weeks, a weekly lag k (weeks before the target) is observable only if $k \geq H$. So the “copy recent weeks” baseline must use the four *most recent available* weeks $\{H, \dots, H+3\}$, which grow stale as H grows: at $H=8$ the freshest usable observation is two months old and may sit on the far side of a sub-annual seasonal boundary (summer vs. back-to-school). All results here are pre-COVID *steady state* (test 2019), so this is the everyday behaviour, not a regime-break artifact.

To separate staleness from irreducible error we add a horizon-*independent* floor: a *centered oracle* that averages the weeks $\{\pm 1, \pm 2\}$ around the target (it peeks at the future, so it has perfect concurrent level and no staleness). Its RMSE is 38.1 veh/h ($R^2=0.946$) at *every* horizon. The gap between the naive baseline and this floor is exactly the prize a better level/trend model could win. Figure 1 and Table 1 trace it over $H = 1, \dots, 12$ weeks, alongside a simple recoverable model (an optimal blend with a prior-year, calendar-aligned *annual anchor*).

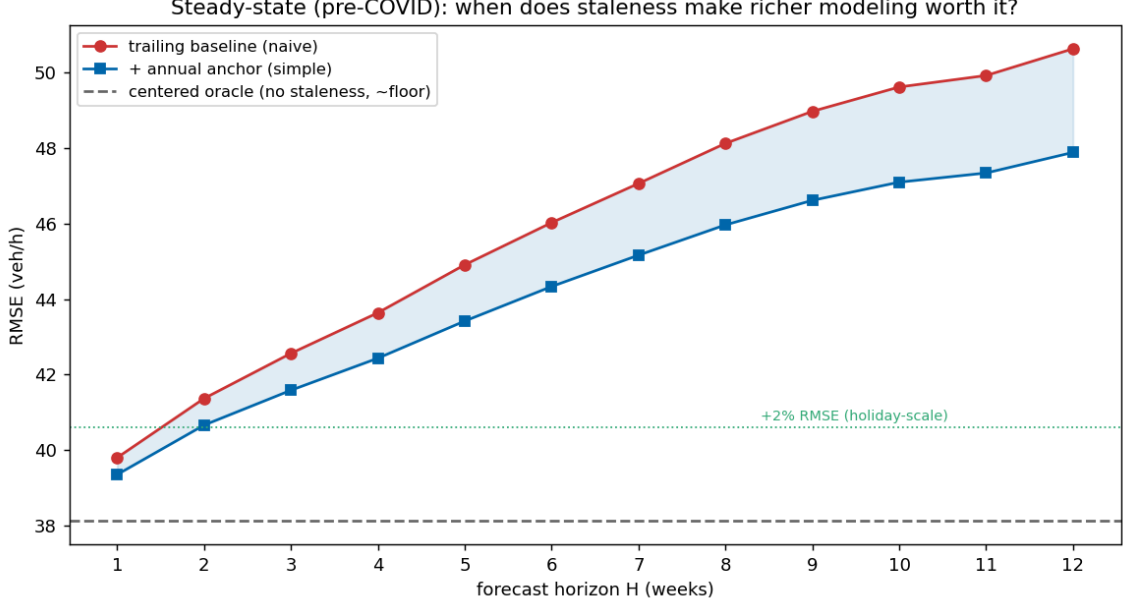


Figure 1: Steady-state RMSE vs. horizon. The naive baseline (red) pulls away from the no-staleness floor (dashed) roughly linearly; the shaded band is the recoverable prize. A simple annual anchor (blue) recovers about half of it; the rest needs genuine trend modeling. “Interesting” (richer modeling worth a holiday-scale margin) begins around $H=3-4$ weeks.

Table 1: Steady-state horizon sweep (test 2019). Gap-to-floor is the share of RMSE attributable to level staleness – the target for richer modeling.

Horizon	Naive RMSE	Gap to floor	Optimal annual weight
1 week	39.8	4%	0.20
2 weeks	41.4	8%	0.23
4 weeks	43.6	13%	0.27
8 weeks	48.1	21%	0.34
12 weeks	50.6	25%	0.39

The shape of the curve is the answer: there is **no sharp knee**. Each extra week of horizon ages the anchor by one week and adds roughly constant level-drift variance, so the gap-to-floor grows $\sim$ linearly – 4% $\rightarrow$ 13% $\rightarrow$ 25% over 1–12 weeks. Reading off a threshold: below ~ 2 weeks the naive rule is within noise of the floor (solved); at $H \approx 3-4$ weeks the recoverable gap reaches the scale of gains we already judged worth building (the holiday model’s $\sim 2\%$ RMSE), so a level/trend model starts to earn its keep; by 8–12 weeks a quarter of the error is staleness and the naive rule is clearly leaving accuracy on the table. The annual anchor – nearly worthless at one week (Note 02’s YoY result) – recovers an increasing share as it becomes horizon-independent leverage, but only about *half* the gap; closing the rest needs an actual model of *where we are in the year* and of the slow trend, not a copy of recent weeks.

3 The trend band: 2017–2021

Zooming out to five years (Figure 2) shows the decomposition starkly. The network-mean level is nearly flat through early 2020 – a slow, gently growing trend with a tight seasonal oscillation on top – and then falls off a cliff at the March 2020 shelter-in-place order, partially recovering through 2021.

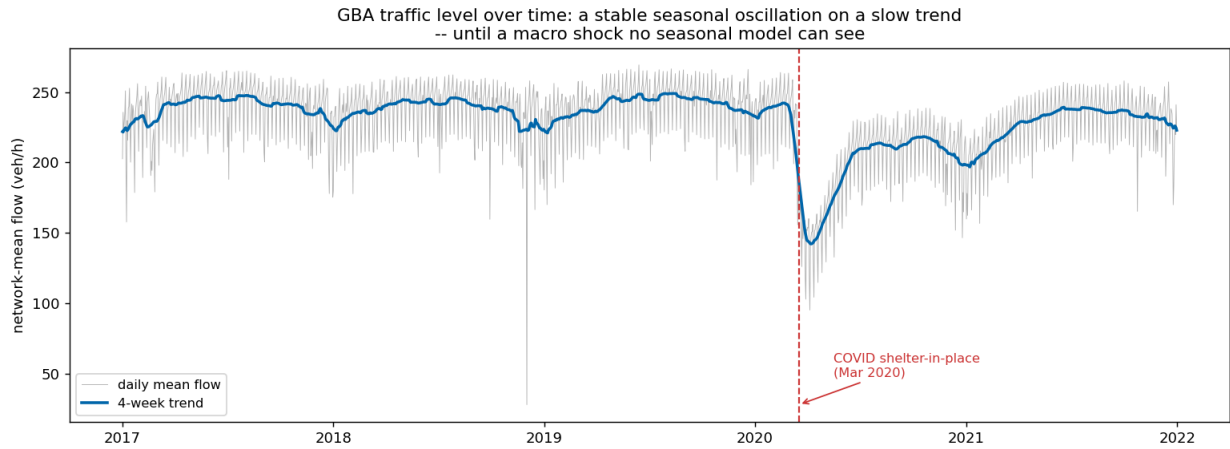


Figure 2: Network-mean GBA flow, 2017–2021. A stable seasonal oscillation rides a slow trend – until a macro shock (COVID) that no seasonal model can anticipate.

Meanwhile the weekly *shape* barely moves. Cross-year correlations of the normalized weekly profile against 2019 are 0.9994 (2017), 0.9994 (2018), 0.9953 (2021) – essentially frozen – with mean levels 239, 237, 241, 205, 228 veh/h for 2017–2021 (Figure 3). Even 2020, the exception, mostly drops the *level* (and flattens the commute peaks under remote work) while preserving the shape (0.990). The lesson: **the seasonal band is boringly stable; essentially all the inter-temporal action lives in the level/trend band**, and that band is where macro forces enter.

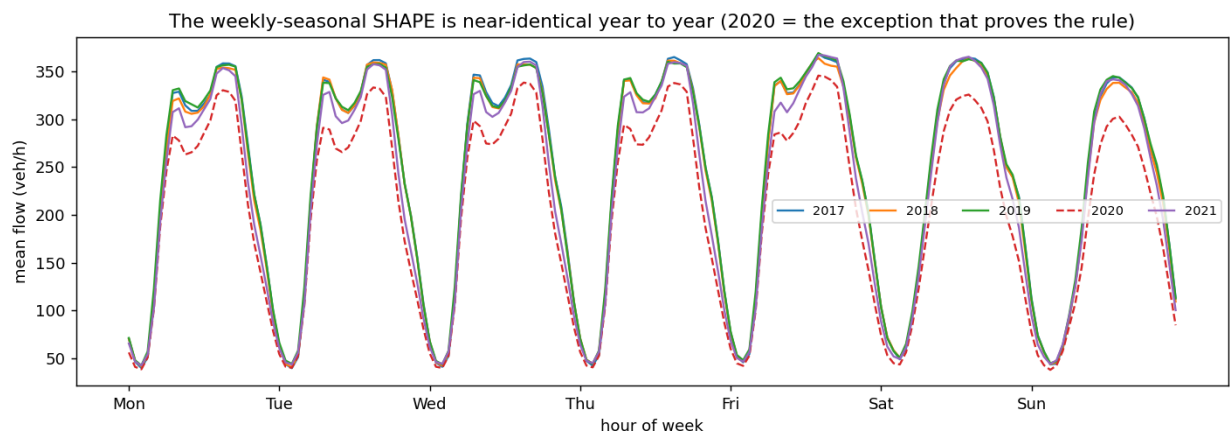


Figure 3: The weekly-seasonal shape is near-identical year to year; 2020 (dashed) drops the level and flattens the commute peaks but keeps the shape.

3.1 The tail: regime breaks

The COVID break is worth one paragraph not as the main story but as the *extreme tail* of the same level band. A trailing one-week-ahead forecaster has a *median* 2020 weekly-aggregate MAPE of 3.6% – normal! – because once the new regime persists for four weeks the trailing window re-adapts. But *at the transition* it fails catastrophically: the weeks of March 17 and March 24, 2020 carry weekly MAPEs of 58% and 60% (a $16\times$ blowup), decaying back over ~ 4 –6 weeks. Seasonal models do not see structural breaks; they only recover from them. Such breaks are rare one-offs, but they share a mechanism with the everyday onramp of §2: both are the *level/trend band* asserting itself, gently as horizon grows and violently when a macro shock hits. At horizons or in regimes where that band dominates, traffic forecasting *is* trend forecasting – coupled to employment, fuel prices, population, and remote-work structure – and inherits their difficulty; this is where the “sunrise” analogy gives way to the “stock price” one.

4 Scope for richer statistical methods

Does this leave room for methods beyond hand-crafted seasonal rules? The honest answer from Notes 01–03: **model sophistication earns its keep when the objective or the available information changes – not on the original task.** On one-week-ahead point forecasting of flow we proved the simple model sits at the ceiling and ML on residuals adds nothing (Note 02). But four shifts open real scope:

1. **Longer horizons $\Rightarrow$ trend modeling.** Forecasting the *level* months–years out is a genuine, structured problem. State-space / structural time-series models (e.g. Bayesian structural time series, dynamic linear models), Gaussian processes, and models ingesting *exogenous macro covariates* (employment, fuel price, population, remote-work indices) are the right tools – and uncertainty grows with horizon, so they should be *probabilistic*.
2. **Distributional targets $\Rightarrow$ uncertainty modeling.** Everything here was point accuracy, but much traffic value lives in the tails: P (severe congestion), calibrated prediction intervals, risk. The residual is strongly heteroscedastic (incident-driven heavy tails at peaks and transitions), which point rules ignore. Quantile regression, conditional density / distributional forecasting, and GARCH-type variance models have clear scope – not for better means, but for calibrated risk.
3. **Robustness to regime change $\Rightarrow$ adaptive models.** The COVID failure motivates Markov-switching models, online change-point detection, and structural-time-series with explicit level-shift components – which detect and adapt to breaks rather than merely recovering from them. The payoff is robustness, not average-case accuracy.
4. **Richer information $\Rightarrow$ learned global / spatial models.** Once genuinely new signal is added – weather, events, network topology – models that capture nonlinear interactions and pool across sensors (gradient boosting, hierarchical Bayes, spatiotemporal graph networks) become appropriate. Spatial models specifically help *short*-horizon incident nowcasting and missing-sensor imputation, where cross-corridor structure is real (it is not, at the week-ahead horizon – Note 02).

What we found when we tried. We ran this ladder on the weekly level at $H=4$ –8 weeks: a recency rule (trailing mean of the most recent observable weeks), a fitted classical model (deseasonalized Holt), a *pooled* panel AR with coefficients shared across all 2,352 sensors, and a trained

global gradient-boosted model. On *endogenous* information the simplest rule is again at the ceiling: the only thing that beats trailing is a one-parameter blend toward the prior-year value, while the pooled AR (even with a smooth low-variance Fourier seasonal) and the GBM *underperform* it – on a series 96% predictable from recency, “estimate + tiny shrinkage” beats reconstructing the level from components. One metric caution: the weekly-level R^2 (~ 0.96) is flattered by the annual cycle in its denominator; the honest measures (RMSE, which grows +47% from one to six weeks, and skill over seasonal-naïve) show the horizon effect that R^2 hides.

Intrinsic or data-limited? That verdict, though, rests on three pre-COVID years of *one* region – and the level band is exactly where more data could change it. Cross-regional co-movement (a common statewide level factor), multi-year cycle/trend structure, and hierarchical shrinkage of noisy per-sensor levels are all signals a larger panel could exploit and a single short region cannot. The decisive cheap test is whether the deseasonalized level co-moves across regions; if so, a forecastable common factor exists and pooling is warranted. So we hold the “rules win” result as *sample-limited*, not universal.

The unifying principle is the lesson of this project in miniature: **complexity should follow information and objective, not precede them**. The reason XGBoost lost on Note 02’s task – and pooled models lose here – was an information gap, not a modeling gap; change the task (longer horizon, distributional, regime-robust) or the inputs (known-ahead covariates, more regions and years) and the calculus flips. The most promising concrete next steps are therefore (a) the cross-regional co-movement test and a pooled/hierarchical level model, (b) known-ahead exogenous covariates (events, planned closures, calendar), and (c) distributional forecasting for congestion risk – each targeting value the seasonal baseline structurally cannot reach.

Reproducibility. Figures and numbers from `src/: horizon_sweep` (§3, fig. 13, the steady-state onramp), `multiyear_trend` (§4, figs. 11–12, and the COVID-transition quantification inline), with `multiweek_horizon` as the coarse precursor.